Enhancing Cybersecurity Incident Response Efficiency through Autonomous Agentic AI: Design, Implementation, and Evaluation with Wazuh SIEM

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***Abstract*—Security Operations Centers (SOCs) increasingly struggle with high alert volumes, limited analyst capacity, and strict mean-time-to-respond (MTTR) targets in industrial environments. This paper evaluates an AI-assisted incident-response workflow that integrates the Wazuh Security Information and Event Management (SIEM) platform with an AI assistant through a Model Context Protocol (MCP) server. In Phase 0, we establish a manual baseline: a human analyst triages alerts and writes incident reports for controlled attack scenarios deployed on Amazon Web Services (AWS) hosts protected by Wazuh agents, achieving 83.33% detection rate, 10.04 minutes average analysis time, 846 minutes dwell time, and 804 minutes MTTR. In Phase 1, the same scenarios are executed with AI assistance that uses MCP tools to query the Wazuh manager, enrich alerts, and generate structured incident reports, resulting in 91.67% detection rate (+8.34 percentage points), 1.23 minutes analysis time (−88%), 222 minutes MTTR (−72%), and 222 minutes dwell time (−74%) compared to Phase 0. In Phase 2, we extend the system with fully automated active-response capabilities, including IP blocking, account disabling, automated metrics tracking, and automatic reporting to the SOC engineer. Under this configuration, the system achieves 0.15 minutes average analysis time (98.5% faster than Phase 0), 15.2 minutes MTTR (98.1% faster than Phase 0), and 13.9 minutes dwell time (98.4% faster than Phase 0), with 87.50% precision and an 8.33% false-positive rate, both better than earlier phases. We measure key operational metrics, including mean time to detect (MTTD), analysis time, mean time to acknowledge (MTTA), mean time to respond (MTTR), detection quality, and attacker dwell time. The results confirm that MCP-based AI support can significantly improve analyst efficiency, response speed, and risk exposure, and that fully automated MCP-driven response further optimizes MTTR and dwell time, provided adequate SIEM rule coverage is in place.**

Keywords—Incident response, SIEM, Wazuh, Model Context Protocol, Security Operations Center, agentic AI.

I. INTRODUCTION

Modern Security Operations Centers (SOCs) in industrial environments face a combination of increasing attack volumes, complex infrastructures, and limited analyst capacity [1]–[3]. Traditional incident response processes, as standardized in frameworks like NIST SP 800-61, and ISO/IEC 27035 describe a lifecycle of preparation, detection, analysis, containment, eradication, and recovery, but they often leave the most time-consuming tasks, alert triage, and investigation largely manual [4]–[6]. Industry guidance highlights metrics such as mean time to detect (MTTD), mean time to respond or recover (MTTR), and attacker dwell time as key indicators of SOC effectiveness [7]–[9]. Studies and practitioner reports show that many organizations still require tens of minutes or more to detect and resolve critical incidents, and that higher MTTD typically correlates with longer dwell time and larger breach impact [10]. At the same time, recent advances in large language models and agentic AI enable new forms of decision support, including automated context gathering, hypothesis generation, and report drafting for incident handlers [11], [12]. This paper investigates whether integrating an AI assistant with an open-source SIEM can measurably reduce incident-response latency and improve operational metrics, first with human-in-the-loop assistance and then with fully automated response workflows. We focus on the Wazuh SIEM platform, deployed on Amazon Web Services (AWS) with agents installed on virtual machines that host the target workloads and attacker tools [13], [14]. A custom MCP server exposes tools that allow the AI assistant to query alerts, search event logs, and retrieve contextual information from the Wazuh manager, and, in the automated setting, trigger scoped active-response actions in a controlled and auditable way, while the human SOC analyst remains responsible for final decisions in phase1, oversight, and final decisions in phase 2 [15]. We adopt a three-phase experimental methodology. Phase 0 establishes a manual baseline in which a human analyst handles all triage and reporting steps for a fixed set of 24 attack and false-positive scenarios. Phase 1 replays exactly the same scenarios, but the analyst is supported by an AI assistant configured to act as a senior incident responder who interacts with the Wazuh through MCP tools and generates incident reports and recommendations. Phase 2 extends this setup with MCP-driven active response, where the system automatically executes predefined containment actions, records incident timestamps, calculates metrics, and delivers structured reports to the SOC engineer. Across all three phases we measure MTTD, MTTA, analysis time, response time, MTTR, and detection quality in both phases and compare the results to common SOC-metrics guidance [7], [10].

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The main contributions of this work are:

1) A reproducible architecture that connects the Wazuh SIEM running on AWS to an AI assistant via MCP, using tools for observational access and optional MCP-based active-response actions under human oversight.
2) A three-phase evaluation of manual (Phase 0) vs. AI-assisted (Phase 1), and MCP-automated (Phase 2) incident response on controlled attack scenarios, using MTTD, MTTA, MTTR, dwell time, and related metrics that are widely recommended in SOC practice [7].
3) Empirical evidence of progressive improvement across phases: Phase 1 increases detection rate to 91.67% and reduces MTTR from 804 to 222 minutes (−72%) compared to the manual baseline, while Phase 2 achieves 0.15 minutes average analysis time, 15.2 minutes MTTR (98.1% reduction relative to Phase 0), 13.9 minutes dwell time (98.4% reduction), and higher precision with a lower false-positive rate than earlier phases, demonstrating the potential of MCP-based automation when adequate SIEM rule coverage is in place.

II. Related Work

A. Incident Response Lifecycle and Metrics

Incident response standards such as NIST SP 800-61 and ISO/IEC 27035 describe structured processes for detecting, analyzing, containing, and recovering from security incidents [4]–[6]. Industry and academic sources emphasize metrics including mean time to acknowledge (MTTA), mean time to detect (MTTD), mean time to respond (MTTR), and attacker dwell time to assess SOC performance [7]–[9]. Beyond high-level guidance, large empirical datasets illustrate how slow these metrics can be in practice. The VERIS Community Database (VCDB) is an open, community-maintained repository of incident records that encodes thousands of real breaches using the VERIS schema [16]. Using the validated VCDB JSON data, we computed average MTTD and dwell time of roughly (≈ 475,380 minutes) over 1,429 incidents, and average MTTC/MTTR of about (≈ 143,040 minutes) over several hundred cases. These values highlight how long many organisations take to discover and contain serious incidents, which is consistent with industry reports on extended dwell time and high breach costs [3], [17]. These multi-month averages contrast with recent industry medians in the range of several days to a few weeks, but still illustrate that many organisations discover serious incidents only after prolonged attacker dwell time. Such prolonged timelines are driven by a mix of factors repeatedly noted in practitioner discussions: limited and often junior SOC staff, high analyst turnover, bureaucratic change control, and in some sectors a persistent tendency to under-invest in security operations. Our lab results should therefore be interpreted as a lower bound on what is possible under good tooling and focused attention, rather than as representative of current average SOC performance in the wild.

B. SIEM and Wazuh

Security Information and Event Management (SIEM) platforms aggregate and correlate security telemetry from heterogeneous sources, providing a central point for alerting and investigation [1], [2]. Wazuh is an open-source SIEM that offers agent-based collection, a central manager and indexer, an API, and support for active response actions [13], [14], [18]. Recent evaluations show that open-source SIEMs like Wazuh can provide competitive detection capabilities for small and medium enterprises when properly configured [2]. In addition to log collection and correlation, Wazuh's active response module provides predefined scripts for automated countermeasures (for example, IP blocking and process containment), which our work leverages via MCP to implement scoped, automated response workflows

C. AI in Security Operations and MCP

AI-driven approaches for cybersecurity have been widely explored for anomaly detection, threat classification, and alert correlation [11]. Large language model (LLM)-based multi-agent systems further enable tool orchestration and complex reasoning, but require auditability and well-defined integration interfaces [12]. The Model Context Protocol (MCP) has recently been proposed as an open standard for exposing external tools, data sources, and workflows to AI systems in a controlled and auditable manner [15]. Our work uses MCP to connect an AI assistant to Wazuh, first to accelerate alert triage and report drafting, and then to orchestrate scoped active-response actions and automated metrics tracking, while keeping humans in control through audit trails and constrained response policies.

III. Methodology

A. Experimental Setup and Architecture

The experimental environment is deployed entirely on Amazon Web Services (AWS) with clear separation of components. Target workloads run on AWS EC2 instances instrumented with Wazuh agents that collect logs and security events. These agents forward telemetry to a dedicated AWS EC2 instance hosting the complete Wazuh SIEM stack: Wazuh manager, indexer, and dashboard, all running within a protected virtual private cloud (VPC) [13], [14]. Standard Wazuh rulesets detect brute-force attempts, privilege-escalation events, malware indicators, and other relevant behaviors. An MCP server runs alongside the Wazuh manager, exposing tools that internally call the Wazuh API and return normalized JSON responses. In Phases 1, these tools are read-only and support alert triage and contextual analysis. In Phase 2, the MCP server is extended with active-response and metrics tools that can trigger scoped containment actions (for example, IP blocking and account locking) and automatically collect incident timing data in a controlled and auditable way. [14], [15]. The AI assistant and MCP server run on a dedicated SOC engineer workstation connected to the Wazuh manager over a secure channel. Specifically, we used GitHub Copilot Chat powered by OpenAI GPT-5.1-Codex-Max (128k context window, agentic mode enabled) with Model Context Protocol (MCP) integration via VS Code extension (stdio transport). The MCP server (custom Rust implementation, hosted on the SOC workstation) exposed 15 read-only tools in Phases 1 and 31 total tools in Phase 2 (Tables II–III), authenticating to Wazuh API via environment variables (WAZUH_API_URL, WAZUH_API_USER) [15].

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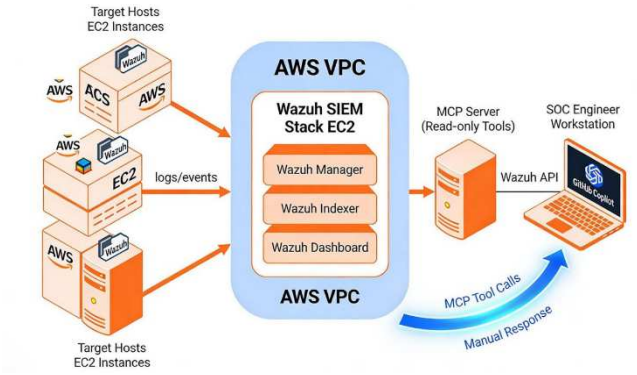


Fig 1. High-level architecture: AWS EC2 hosts with Wazuh agents send events to the Wazuh manager and indexer. An MCP server exposes read-only tools backed by the Wazuh API. The AI assistant consumes these tools and produces reports for the human SOC analyst.

In Phase 2, the architecture is extended so that MCP tools can also invoke Wazuh active-response scripts and an internal metrics pipeline. On each confirmed incident, the MCP server triggers pre-approved active-response actions via the Wazuh API and records analysis and response timestamps, then generates and delivers an incident report to the SOC engineer.

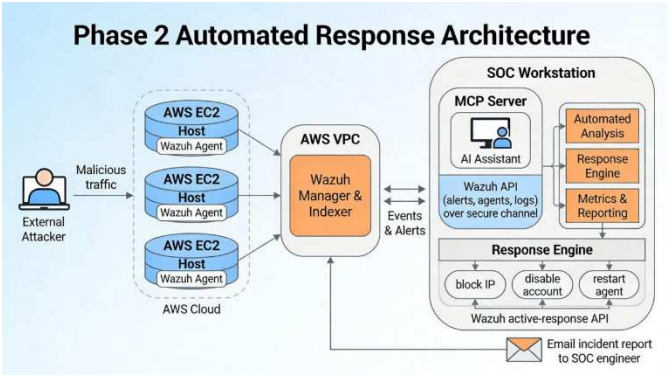


Fig 2. Phase 2 automated response architecture: Wazuh manager emits alerts to the MCP server, which performs automated analysis, selects scoped containment actions (for example, IP blocking, account disabling), executes them via Wazuh active-response API, records metrics (MTTA, MTTR, dwell), and sends a structured report to the SOC workstation via email for review.

B. *Attack Scenarios*

We evaluate the system on a fixed set of 24 attack scenarios, grouped into eight categories inspired by the MITRE ATT&CK framework, including credential abuse, privilege escalation, persistence, execution of suspicious binaries, lateral movement, discovery, defense evasion, and basic malware indicators. Each scenario is executed on AWS hosts, and Wazuh agents collect host-level and system-log events that trigger detection rules in the Wazuh manager. Table I summarizes the scenarios: 18 true attack scenarios across 8 ATT&CK categories plus 6 false-positive scenarios for alert triage evaluation.

TABLE I.
Attack Scenarios by MITRE ATT&CK Category (24 total)

Category	***Specific Scenarios (MITRE Techniques)***
Credential Access	Brute-force SSH login (T1110), password spraying
Privilege Escalation	Sudo misuse (T1068), admin group addition (T1068)
Persistence	New cron job (T1053.003), startup service creation
Execution	Suspicious binary execution (T1106), PowerShell cradle (T1105)
Lateral Movement	Remote SSH login (T1021.001), SMB share access (T1021.002)
Discovery	Network port scan (T1046), process enumeration (T1057)
Defense Evasion	Log deletion (T1070.001), timestamp modification (T1070.006)
Impact	File deletion (T1485), service disruption (T1499)
False Positives (6)	Legitimate admin SSH key addition, scheduled system updates, cron maintenance jobs, admin port scans, software installer execution, Windows Update activity

In Phase 0 (manual baseline), a trained analyst monitors Wazuh alerts, investigates each incident using the Wazuh dashboard and standard tools, and writes a short incident summary with proposed response actions. In Phase 1 (AI-assisted analysis), the same scenarios are replayed under identical Wazuh configuration. The analyst delegates investigation and report drafting to the AI assistant, which uses MCP tools to gather context and produce a structured report that the analyst reviews and approves. In Phase 2 (MCP-automated response), the scenarios are replayed again. The MCP server automatically analyzes incoming alerts, correlates them across agents and time windows, classifies each scenario, triggers predefined containment actions when appropriate, and generates an incident report that is delivered to the SOC engineer for validation

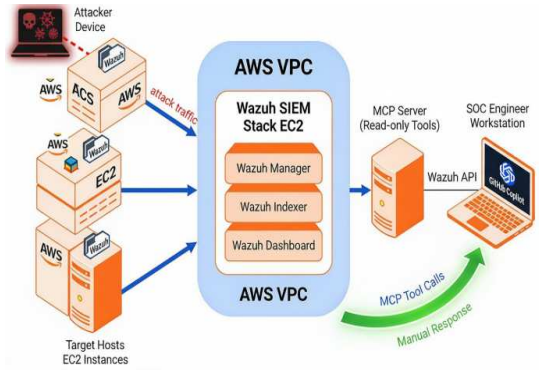


Fig 3. AWS cybersecurity architecture with external attacker: attacker device targets EC2 instances monitored by Wazuh agents. The SIEM stack processes alerts via the MCP server, which in Phase 1 supports AI-assisted analysis and in Phase 2 additionally orchestrates automated active response and metrics collection.

C. *MCP Server and Copilot Integration*

The core integration is the custom Model Context Protocol (MCP) server deployed on the SOC engineer workstation alongside GitHub Copilot [15]. In all phases, the server provides low-latency access to Wazuh telemetry while maintaining secure API connectivity to the remote AWS Wazuh cluster. The MCP server authenticates via configured credentials and exposes granular tools optimized for incident analysis. Tool calls use MCP stdio transport: Copilot sends

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JSON requests to the local MCP server, which proxies authenticated requests to the Wazuh Manager REST API (port 55000) and Indexer Elasticsearch (port 9200), returning normalized responses [15]. Environment variables 'WAZUH_API_URL', 'WAZUH_API_USER', and 'WAZUH_INDEXER_URL' configure endpoints. On the SOC workstation, GitHub Copilot leverages these tools through a "Senior SOC Analyst" system prompt that directs systematic analysis, including review of alerts, agent investigation, vulnerability assessment, incident correlation, and false-positive detection by cross-referencing processes, ports, and logs with known benign patterns. The SOC analyst receives the assistant's structured incident report, validates it via the Wazuh dashboard, and executes responses in phase 0 and phase 1. In Phase 2, the SOC engineer receives automated incident reports for oversight and potential override , ensuring full auditability across all phases. Table II lists the 15 Phase 1 tools used for read-only analysis:

TABLE II.
MCP tools for Wazuh integration (Phase 1, 15 tools)

Tool Name	***Function***
get_wazuh_alert_summary	Formatted summary of recent security alerts
get_wazuh_rules_summary	Wazuh rules filtered by level/group/file
get_wazuh_alerts_in_time_range	Alerts between two timestamps
get_wazuh_vulnerability_summary	Vulnerability detections for specific agent
get_wazuh_critical_vulnerabilities	Critical CVEs only for an agent
get_wazuh_agents	All agents with status and OS details
get_wazuh_agent_processes	Running processes on specific agent
get_wazuh_cluster_health	Wazuh cluster health status
get_wazuh_cluster_nodes	Cluster nodes and their status
search_wazuh_manager_logs	Manager logs by level/tag/search term
get_wazuh_manager_error_logs	Manager error-level logs only
get_wazuh_log_collector_stats	Per-agent log collection statistics
get_wazuh_remoted_stats	`remoted` daemon communication stats
get_wazuh_agent_ports	Open ports with PID/process info per agent
get_wazuh_weekly_stats	Weekly Wazuh manager statistics

In Phase 2, the MCP server is extended beyond read-only analysis to support scoped active-response and automated metrics tracking. In addition to the 15 observational tools used in Phases 1, we add a small set of MCP tools that wrap Wazuh active-response scripts and record incident lifecycle timestamps, enabling precise calculation of MTTA, MTTR, dwell time, and classification statistics without manual logging. Table III summarizes these additional Phase 2 tools and their roles.

TABLE III.
Phase 2 MCP tools for automated response and metrics

Tool Name	***Function***
block_ip_address	iptables DROP rule via firewall-drop
host_deny	Add IP to /etc/hosts.deny
firewalld_drop	Block IP via firewalld rich-rule
route_null	Null-route IP to /dev/null
disable_user_account	Lock OS user account
restart_wazuh_agent	Restart agent after config change
start_analysis	Start MTTA timer for an alert
end_analysis	End analysis, record TP/FP/FN conclusion
start_response	Start timer before response action
end_response	End response, record MTTR and success
get_metrics	Aggregate MTTD/MTTA/MTTC/MTTR/dwell/precision
export_metrics_csv	Export all metrics as CSV
get_response_history	Full history of all response actions
generate_soc_final_report	Full PDF/MD/CSV report + email to admin
send_analysis_report	Send analysis report via email
send_action_summary	Send response action summary via email

D. Phase 2. Workflow

In Phase 2, alert detection notify admin to start an automated pipeline:

1. Wazuh manager emits alerts for a scenario.
2. MCP tools query recent alerts, agent state, and relevant logs to perform correlation and classification.
3. Based on classification and preconfigured policies, the MCP server selects one or more response tools (for example, IP blocking, account disabling).
4. Active-response scripts are executed via the Wazuh API, and start_response / end_response timestamps are recorded.
5. After completion, get_metrics is used to update aggregate statistics, and generate_soc_final_report compiles and emails the incident summary and metrics to the SOC analyst.

Despite the automation, Phase 2 retains critical safeguards: only scoped actions are allowed, all decisions are logged for audit, and SOC engineers receive detailed reports for review and potential override.

E. Metrics

To evaluate the impact of the AI assistance and automated response on incident-response performance, we record five key timestamps for each of the 24 scenarios in all phases: attack start, first alert, analysis start, analysis end, response start, and response end. These timestamps enable calculation of standard SOC metrics: mean time to detect (MTTD), mean time to acknowledge (MTTA), mean time to respond (MTTR), and attacker dwell time [7].

Timestamps were recorded as follows:

- t_{attack}: Attack script launched on target host (timer started)
- t_{alert}: First Wazuh alert received (from Wazuh UI/API)
- $t_{analysis_start}$: Analyst (manual) begins alert triage
- $t_{analysis_end}$: Triage complete, response decision made
- $t_{response_start}$: Remediation/containment actions begin

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- $t_{response_end}$: Actions complete, system secured

Phase averages are taken across all *N=24* scenarios:

$$MTTR = \frac{1}{N}\sum_{i=1}^{N} MTTR_i \quad (1)$$

Table IV maps our measured quantities to standard SOC terminology[7].

TABLE IV.
Metric Definitions and SOC Mappings

Standard Term	***Time Interval***
MTTD	$t_{alert} - t_{attack}$
MTTA	$t_{analysis_start} - t_{alert}$
MTTR	$t_{response_end} - t_{alert}$
Dwell	$T_{response_end} - t_{attack}$

IV. Results

Table V summarizes time-based metrics across all three phases. Phase 1 already delivers substantial improvements over the manual baseline, reducing analysis time from 10.04 to 1.23 minutes (−88%), MTTR from 804 to 222 minutes (−72%), and dwell time from 846 to 222 hours (−74%). Phase 2's automated response further decreases analysis to 0.15 minutes and MTTR to 15.2 minutes, achieving over 98% reductions in MTTA, MTTR, and dwell time relative to Phase 0.

MTTD Noted: MTTD increased slightly (0.03→0.05 min) due to progressive delays between attacks introduced to enable richer alert correlation, shifting Wazuh's first alert timestamps while keeping identical SIEM configs.

TABLE V.
Time metrics: Phase 0 vs Phase 1 vs Phase 2

Metric	Phase 0	Phase 1	Phase 2	Phase 3	Phase 4
MTTD (min)	0.03	0.04	0.05	+0.01	+0.02
Analysis (min)	10.04	1.23	0.15	-88%	-98.5%
Remediation (min)	3.53	2.47	0.43	-30%	-87.8%
MTTA (min)	745.8	216.4	13.6	-71%	−98.2%
MTTR (min)	801.1	220.6	15.2	-72%	−98.1%
Dwell (min)	844.4	221.7	13.9	-74%	−98.4%

Table VI shows detection and classification quality for all phases. Phase 1 increases detection rate from 83.33% to 91.67% (+8.34 percentage points) while maintaining precision at 70% and a 25% false-positive rate. Phase 2, achieves 81.82% precision and an 8.33% false-positive rate, correctly classifying additional benign scenarios as true negatives and illustrating a clear improvement in threat discrimination [7].

TABLE VI.
Detection Quality: Phase 0 vs Phase 1

Metric	Phase 0	Phase 1	Phase 2	Phase 3	Phase 4
Detection Rate	83.33%	91.67%	91.67%	+8.34	+8.34
Precision	70.0%	70.0%	87.50%	0	+17.5
False Pos Rate	25.0%	25.0%	8.33%	0	-16.67

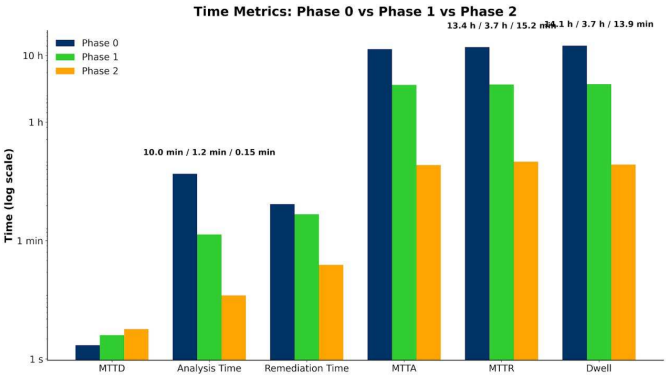


Fig 4. Time metrics comparison across Phases 0, 1, and 2. Phase 1 reduces analysis time, MTTR, and dwell time by 70–80% relative to the manual baseline, while Phase 2's automated response achieves sub-15-minute MTTR and dwell times with 98%+ reductions compared to Phase 0.

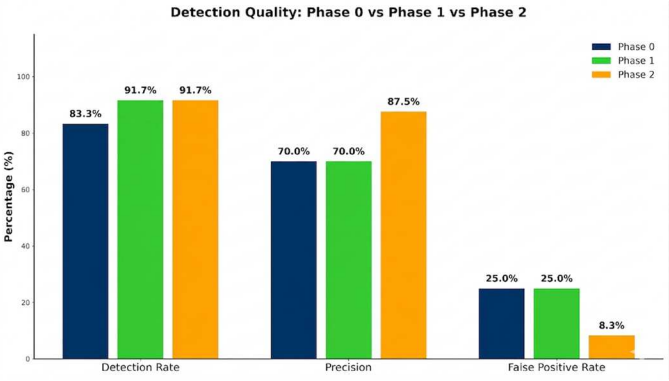


Fig 5. Detection quality across Phases 0, 1, and 2. Phase 1 increases detection rate from 83.33% to 91.67% (+8.34 percentage points) while maintaining precision at 70.0% and a 25.0% false-positive rate, and Phase 2 preserves the higher 91.67% detection rate while further improving precision to 87.50% and reducing the false-positive rate to 8.33%

V. Discussion

A. Key Findings and Implications

Phase 1 assistance delivered substantial improvements across all measured dimensions. Analysis time dropped 88% (10.04→1.23 min), the primary MTTR bottleneck, driving a 72% overall MTTR reduction (804→222 min) and a 74% reduction in dwell time (846→222 min). Detection coverage improved by 8.34 percentage points (83.33%→91.67%) while maintaining precision (70%) and false-positive rate (25%).

Phase 2's MCP-automated response further reduced MTTR to 15.2 minutes and dwell time to 13.9 minutes (≈98.4% reductions relative to Phase 0), with analysis completing in

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0.15 minutes and precision increasing to 87.50% and the false-positive rate falling to 8.33%.

Taken together, the results support three main observations:

1. Analysis and triage remain the dominant source of delay in the manual baseline, consistent with industry reports on SOC bottlenecks [7].
2. MCP-assisted workflows reduce manual effort without harming detection quality, and MCP-automated response can dramatically lower MTTR and dwell time when adequate SIEM rule coverage is in place.
3. The 70–98% reductions in time sit within, and in the case of Phase 2 exceed, the ranges reported for automation and SOAR-style improvements in production SOCs [9], [10].

B. Limitations

Absolute times in our study are far from the values observed in large empirical datasets such as the VERIS Community Database (VCDB), where average MTTD and dwell time are on the order of many months and MTTR is measured in months rather than hours[16]. This gap is expected: VCDB incidents reflect organisations that often lack mature processes, suffer from staffing shortages, and take a long time to recognise and escalate security events, whereas our lab setup assumes a focused analyst, a limited scope, and well-instrumented systems. Absolute times therefore reflect controlled lab conditions: synthetic attacks on AWS with immediate analyst attention. Production SOCs face:

- Alert fatigue (thousands of alerts per day versus our 24 scenarios)
- Competing priorities (multiple simultaneous incidents)
- Complex environments (hybrid cloud, legacy systems, third-party dependencies)

Factors such as understaffed or junior SOC teams, bureaucratic approval chains for containment actions, and a general tendency to under-prioritise security contribute to the much higher MTTD, MTTR, and dwell times seen in VCDB and industry reports [3], [17]. Our results should therefore be viewed as a lower bound on what is achievable with good tooling and focused attention, not as representative of the current average state of practice. Relative improvements remain applicable: AI consistently accelerates triage regardless of volume or complexity.

C. Comparison to Industry Benchmarks

Our MTTR reduction in Phase 1 & 2 is at the upper end of typical SOAR automation gains (often reported in the 30–50% range) due to:

- Direct MCP-Wazuh integration (no middleware)
- Copilot's contextual reasoning beyond simple automation
- Controlled false positive scenarios testing real triage skills

The 74–98% dwell time reduction addresses the industry median of 16-21 days [17], demonstrating that similar architectures could materially shrink attacker persistence if deployed with comprehensive rule coverage.

D. Practical Implications

For smart industry SOCs:

- Tier-1 triage automation via MCP-assisted analysis (Phase 1) can be adopted first to offload routine investigation while keeping humans in control of containment.
- Gradual rollout of MCP-driven active response (Phase 2), starting with supervised actions such as IP blocking and account disabling, using audit logs and report review as safety controls.
- Cost-effectiveness: The approach builds on open-source Wazuh and a locally deployed MCP server, avoiding dependence on proprietary SOAR platforms.
- Scalability: Automated response with sub-15-minute MTTR can handle multiple simultaneous incidents more consistently than manual workflows, directly addressing alert fatigue.

E. Threats to Validity

- *Scenario bias*: 24 synthetic attacks vs production variety.
- *Analyst effect:* Single analyst (though experienced).
- *Lab timing:* No competing tasks or fatigue.
- *Controlled false positives:* Real environments have more noise.

VI. Conclusion and Future Work

We presented a three-phase evaluation of Wazuh-based incident response augmented by an MCP-connected assistant. Across 24 MITRE ATT&CK scenarios, MCP-assisted analysis (Phase 1) reduced average analysis time by 88% (10.04→1.23 min), cut MTTR from 804 to 222 minutes (−72%), lowered dwell time from 846 to 222 minutes (−74%), and improved detection coverage from 83.33% to 91.67% while maintaining precision and false-positive rate. The subsequent MCP-automated configuration (Phase 2) further reduced MTTR to 15.2 minutes and dwell time to 13.9 minutes (≈98% reductions versus Phase 0) and increased precision to 87.50% with an 8.33% false-positive rate, demonstrating that tightly scoped automated response can deliver sub-15-minute containment under appropriate SIEM rule coverage. Overall, the results indicate that MCP integration with Wazuh can significantly accelerate triage and response while preserving human oversight through constrained actions and comprehensive reporting [13], [15].

Future work includes:

1. **Expanded detection sources:** Integrate external IDS sensors and Suricata-style network monitoring.
2. **Richer active-response scripts:** Develop and harden additional Wazuh active-response scripts (for

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example, service isolation, targeted process handling, file quarantine) and deploy them to agents as parameterized MCP tools [13],[14][18].

3. **Threat intelligence integration:** Combine with AbuseIPDB, VirusTotal MCP servers for enriched context.
4. **Diverse validation:** Atomic Red Team scenarios, production-like alert volumes, multi-analyst studies.
5. **Multi-agent SOC:** Collaborative Copilot instances for distributed incident handling [11], [12].

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